EMERGING TRENDS IN 5G TECHNOLOGY

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ABSTRACT:

Today the internet speed is growing at an unimaginable pace. The abstract of this paper is study based on 5G Technology of Mobile Communication. In telecommunications, 5G is the fifth-generation technology standard for broadband cellular networks, which cellular phone companies began implementing World Wide in 2019 & is the planned successor to the 4G networks which provide connectivity to current cell phones.5G is the term used to describe the next-generation of mobile networks beyond LTE mobile networks.Latency in a 5G network could get as low as 4 milliseconds in a mobile scenario and can be as low as 1 millisecond in Ultra-Reliable Low Latency Communication scenarios.

KEYWORDS:

5G Technology, Generations, Advantages, Limitations, Devices.

INTRODUCTION:

"In the Future, Everything is going to be transformed by 5G. 5G Ultra-Wideband isn't just another iteration of wireless networks.", says Verizon Communications CEO Hans Vestberg. 5G wireless technology is meant to deliver higher multi-Gbps peak data speeds, ultra-low latency, more reliability, massive network capacity, increased availability, and a more uniform user experience to more users. Higher performance and improved efficiency empower new user experiences and connects new industries.5G Ultra-Wideband will eventually allow even more technology to connect, enabling the Internet of Things on a truly massive scale.

GENERATIONS OF NETWORKS:

The journey of cellular networks from 1G to 5G just took more than 40 years.A cellular network or mobile network is a communication network where the link to and from end nodes is wireless. The changes happened during evolution of networks were

- Cell phones have become smaller.
- Downloads speeds have become faster.
- Surfing the internet with phones became common.
- Text Messaging has come.

The evolution of cellular networks are

I. 1G Era (1987 - 1991)

"It was Phenomenal but was not enough". Engineer Neil Papworth sent the first SMS on December 3rd, 1992, when he wrote "Merry Christmas" on a computer & sent it to the cell phone of Vodafone director Richard Jarvis. It introduced Analog Voice. This was the first generation of cell phone technology . The very first generation of commercial cellular network was introduced in the late 70's with fully implemented standards being established throughout the 80's. It was introduced in 1987 by Telecom (known today as Telstra), Australia received its first cellular mobile phone network utilising a 1G analog system. 1G is an analog technology and the phones generally had poor battery life and voice quality was large without much security, and would sometimes experience dropped calls . These are the analog telecommunications standards that were introduced in the 1980s and continued until being replaced by 2G digital telecommunications. The maximum speed of 1G is 2.4 Kbps. Motorola DynaTAC weighs 2 pounds & allows us to talk for 30 minutes & took roughly 10 hours to charge.

Motorola DynaTAC 3200

Advantages of 1G Network:

- Improve voice clarity
- The network uses the analog signals.
- Reduce noise in the line
- Secrecy and safety to data & voice calls
- Consume less battery power

What went wrong??

- Poor Voice Quality
- Poor Battery Life
- Large Phone size
- No security
- Limited capacity
- Very slow speed
- Poor hand-off reliability
- It makes use of the mobile phone with the analog signal more difficult and this signal are suffer from interference problem.

II. 2G Era (1991 - 2001)

“A Revolutionary milestone in telecommunication sector.” Following the success of 1G, 2G launched on the Global System for Mobile Communication in Finland during 1991. It transfers a typical file of 40KB in 2 seconds.Cell phones received their first major **upgrade** when they went from 1G to 2G. The main difference between the two mobile telephone systems (1G and 2G), is that the **radio signals** used by 1G network are analog, while 2G networks are **digital** . Main motive of this generation was to provide secure and reliable communication channel. It implemented the concept of **CDMA** and **GSM** . Provided small data service like SMS and mms. Second generation 2G cellular telecom networks were commercially launched on the GSM standard in Finland by Radiolinja (now part of Elisa Oyj) in 1991. 2G capabilities are achieved by allowing multiple users on a single channel via multiplexing. During 2G Cellular phones are used for data also along with voice. The advance in technology from 1G to 2G introduced many of the fundamental services that we still use today, such as SMS, **internal roaming** , conference calls, call hold and billing based on services e.g. charges based on long distance calls and real time billing. The max speed of 2G with General Packet Radio Service (**GPRS**) is 50 Kbps or 1 Mbps with Enhanced Data Rates for GSM Evolution (**EDGE**). Before making the major leap from 2G to 3G wireless networks, the lesser-known 2.5G and 2.75G was an interim standard that bridged the gap.

Nokia 3310

Advantages of 2G Network:

- Emit less radio power
- Mobile unit max power is 20W

What went wrong?

- Systems were unable to handle complex data like videos.
- Required strong digital signals to help mobile phones work. If there is no network coverage, digital signals weakened.
- As the generation moved towards 3G, the user requirement was too complex for 2G to satisfy.

III. 3G Era (2001 - 2009)

“Showed us the unending potential of Data yet did not fulfil its own potential.” 3G network was introduced in 1998, but established in early 2000s. Data speeds went up even further with maximum speed of 42mbps. This generation set the standards for most of the wireless technology we have come to know and love. Web browsing, email, video downloading, picture sharing and other **Smartphone technology** were introduced in the third generation. Introduced commercially in 2001, the goals set out for third generation mobile communication were to facilitate greater voice and data capacity, support a wider range of applications, and increase data transmission at a **lower cost** .

The 3G standard utilises a new technology called **UMTS** as its core network architecture - Universal Mobile Telecommunications System. This network combines aspects of the 2G network with some new technology and protocols to deliver a significantly faster data rate. Based on a set of standards used for mobile devices and mobile telecommunications use services and networks that comply with the International Mobile Telecommunications-2000 (**IMT-2000**) specifications by the International Telecommunication Union. One of requirements set by IMT-2000 was that speed should be at least 200Kbps to call it as 3G service.

3G has Multimedia services support along with **streaming** are more popular. In 3G, Universal access and portability across different device types are made possible (Telephones, PDA's, etc.). 3G increased the efficiency of frequency spectrum by improving how audio is **compressed** during a call, so more simultaneous calls can happen in the same frequency range. The UN's International Telecommunications Union **IMT-2000** standard requires stationary speeds of 2Mbps and mobile speeds of 384kbps for a "true" 3G. The theoretical max speed for **HSPA+** is 21.6 Mbps.

Like 2G, 3G evolved into 3.5G and 3.75G as more features were introduced in order to bring about 4G. A 3G phone cannot communicate through a **4G network** , but newer generations of phones are practically always designed to be backward compatible, so a 4G phone can communicate through a 3G or even **2G network** .

Blackberry Curve 9300

Advantages of 3G Network:

- Anywhere access to the internet
- Always online devices
- Faster data transfer rate
- Cheap call rate

What went wrong?

- Messy Architecture
- Demanded 3G compatible handsets
- High power consumption
- Cost of upgrading to 3G devices was too high.

IV. 4G Era (2010 - 2019)

“4G introduced us to the faster world with a greater Data Capacity”. 4G LTE (Long Term Evolution) was a complete redesign and simplification of 3G network architecture, resulting in a significant reduction in transfer latency and thus, increasing efficiency and speeds on the network. A term MAGIC is used in 4G technology,

- M – Mobile Multimedia
- A – Anytime Anywhere
- G – Global Mobility Support
- I – Integrated Wireless Solution
- C – Customized Personal Service

4G is a very different technology as compared to **3G** and was made possible practically only because of the advancements in the technology in the last 10 years. Its purpose is to provide **high speed** , high quality and high capacity to users while improving security and lower the cost of voice and data services, multimedia and internet over IP. Potential and current applications include amended mobile web access, **IP telephony** , gaming services, high-definition mobile TV, video conferencing, 3D television, and cloud computing.

Apple iPhone 11 Pro Max

The key technologies that have made this possible are **MIMO** (Multiple Input Multiple Output) and **OFDM** (Orthogonal Frequency Division Multiplexing). The two important 4G standards are WiMAX (has now fizzled out) and **LTE** (has seen widespread deployment). LTE (Long Term Evolution) is a series of upgrades to existing UMTS technology and will be rolled out on Telstra's existing 1800MHz frequency band. The max speed of a 4G network when the device is moving is 100 Mbps or **1 Gbps** for low mobility communication like when stationary or walking, latency reduced from around 300ms to less than 100ms, and significantly lower congestion. When 4G first became available, it was simply a little faster than 3G. 4G is not the same as **4G LTE** which is very close to meeting the criteria of the standards. To download a new game or stream a TV show in HD, you can do it **without buffering** .

Newer generations of phones are usually designed to be **backward-compatible** , so a 4G phone can communicate through a 3G or even 2G network. All carriers seem to agree that **OFDM** is one of the chief indicators that a service can be legitimately marketed as being 4G. OFDM is a type of digital modulation in which a signal is split into several narrowband channels at different frequencies. There are a significant amount of infrastructure changes needed to be implemented by service providers in order to supply because voice calls in **GSM** , **UMTS** and **CDMA2000** are circuit switched, so with the adoption of LTE, carriers will have to re-engineer their voice call network. And again, we have the fractional parts: **4.5G** and **4.9G** marking the transition of LTE (in the stage called LTE-Advanced Pro) getting us more MIMO, more D2D on the way to IMT-2020 and the requirements of **5G** .

Advantages of 4G Network:

- High speed & capacity
- Tight security network
- Low cost per bit
- Affordable communication system

What went wrong?

- 4G LTE network needs complex hardware.
- 4G use many antennae & transmitters resulting in poor battery life.
- In areas without 4G coverage, consumers downgraded to 3G while still paying the cost of 4G.
- Higher Data Consumption.

V. 5G Era (2019 -Up ahead of)

"5G is more than a generation, it is a promise to wonderland". South Korea was the first country to offer 5G in March 2019. 5G in Canada has already been introduced in major cities.5G looks at us with a new experience in faster data rates, higher connection density, much lower latency**,** among other improvements.5G is not an upgrade to 4G, but is in a league of its own, because it provides us the ability to connect thousands of devices at once giving the user **the true sense of 'real-time'** experience.This means the computing & processing can move to cloud and resulting in smaller IoT devices, reducing the cost of manufacturing & maintenance by optimizing network for the telecom operator.

Samsung Galaxy A90

IoT will mean a lot different with the growth of 5G, as self-driving cars may yet become a reality even in India.Since increasing subscriber base in tier 1 cities is no longer an enticing prospect for Telcos, they will look at generating new streams of revenue with the maturity of MU-MIMO and achieving data speeds of up to 35Gbps.5G technology may use a variety of spectrum bands, including millimetre wave (mm Wave) radio spectrum, which can carry very large amounts of data a short distance.5G mobile technology can usher in new immersive experiences such as VR and AR with faster, more uniform data rates, lower latency, and lower cost-per-bit.5G can enable new services that can transform industries with ultra-reliable, available, low-latency links like remote control of critical infrastructure, vehicles, and medical procedures.

Here are some revolutionary use cases 5G will bring in:

- Fuelling precision agriculture — even the cotton found in a simple dress uses 5G.
- Enabling reliable wireless IoT connectivity at transport hubs
- Essential to the future autonomous vehicles with V2X & smart logistics within our grasp.
- For closer collaborations at the workplace using real time insights & XRs.
- Bringing immersive, virtual customized shopping experiences anywhere to the consumer.
- Driving the next industrial revolution with flexible manufacturing with XR guided execution, smart surveillance, real time supply chain visibility using blockchain & predictive maintenance.
- With real-time asset tracking and efficient delivery using drones.

5G is a generation currently **under development** , that's intended to improve on 4G. **5G** promises significantly faster data rates, higher connection density, much lower latency, among other improvements. Some of the plans for 5G include **device-to-device** communication, better battery consumption, and improved overall wireless coverage. The max speed of 5G is aimed at being as fast as **35.46 Gbps** , which is over 35 times faster than 4G.

Key technologies to look out for: **Massive MIMO** , Millimetre Wave Mobile Communications etc. Massive MIMO, millimetre wave, small cells, **Li-Fi** all the new technologies from the previous decade could be used to give 10Gb/s to a user, with an unseen low latency, and allow connections for at least **100 billion devices** . Different estimations have been made for the date of commercial introduction of 5G networks. Next Generation Mobile Networks Alliance feel that 5G should be rolled out by **2020** to meet business and consumer demands.

Advantages of 5G Network:

- More efficient & effective
- Data bandwidth of more than 1 Gbps or higher
- Finest quality of services
- Can control the PCs by handsets
- It supports multimedia, voice & internet
- It offers high resolution
- Longer battery life

Limitations of 5G Network:

- Development infrastructure need high cost
- Expensive deal - Many of old devices (1G, 2G, 3G, 4G) wouldn't be the component of 5G, so all of them need to be replaced with a new one. The devices are costly.
- Security & privacy yet be to be solved

CONCLUSION:

This paper gives an overview of 5G network. The special attention is paid on generations of cellular technology. Although the 5G concept is still evolving the review reveals emerging common features. Performance enhancements are mainly expected from a combination of network densification, increased spectrum, and enhanced wireless communication technologies. Machine-type of communications will have increasing proportion of the network connections and traffic. Combination of moving networks and ultra-reliable communications truly calls for novel solutions due to strict technical requirements in challenging propagation conditions. Network virtualization, especially in the form of Cloud RAN development will also have a significant role in 5G. Use cases, scenarios and spectrum allocations altogether have so high variability that utmost agility, scalability and reconfigurability is necessary in the integration of the overall 5G system concept. Although there are already some 5G-relevant documents defining the technical specifications, the research on 5G is still at its initial stage. Looking into the future development of computer, network and communication technologies, 5G is expected to be the future architecture of wireless networks aiming to build a virtual, configurable and intelligent mobile communication systems.

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